

Minimal Weak Phase Retrieval: A Cofactor Characterization, an Algebraic Nullity Theorem, and a Correction to the Published $\mathbb{R}^4$ Example

Candidate partial result for arXiv:2110.06868

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Status. Candidate partial result, likely valid, pending human review. The packet gives a full finite characterization of minimal weak phase retrieval and strengthens Problem 9.1 from non-density to containment in a proper algebraic hypersurface, hence nowhere density and Lebesgue nullity. Problem 9.1 itself was already answered in later literature. The packet does not solve the existence question in Problem 9.2. It does show that the six-vector $\mathbb{R}^4$ example cited as motivation for Problem 9.2 is not weak phase retrievable.

1 Source and extracted problems

The source is F. Akrami, P. G. Casazza, A. Rahimi, M. A. Hasankhani Fard, and B. Daraby, *A note on (weak) phase and norm retrievable Real Hilbert space frames and projections*, arXiv:2110.06868 [1]. Section 9, page 14, asks:

Problem 9.1. Show that the weak phase retrievable frames with $2n - 2$ vectors in $\mathbb{R}^n$ are not dense in all frames with $2n - 2$ vectors.

The source then says that examples with $2n - 2$ vectors are known for $n = 3, 4$, and Problem 9.2 asks for such examples for every $n \geq 5$.

Then we have $TW_1 = [e_1 - e_2]$, $TW_2 = [e_2, e_3]$ and $\{TW_i\}_{i=1}^2$ do not norm retrieval in $\mathbb{R}^3$ since by [14] two 2- dimensional subspaces that are not perpendicular, cannot do norm retrieval in $\mathbb{R}^3$.

9. OPEN PROBLEMS

It is known that the phase retrievable frames are not dense in all frames.

Problem 9.1. Show that the weak phase retrievable frames with $2n-2$ vectors in $\mathbb{R}^n$ are not dense in all frames with $2n-2$ vectors.

We have examples of weak phase retrievable frames with $2n-2$ vectors for $n=3,4$ [6].

Figure 1: Problem 9.1 in the source paper, page 14.

Problem 9.2. Show for every $n \geq 5$ that there are weak phase retrievable frames $\{x_i\}_{i=1}^{2n-2}$ for $\mathbb{R}^n$.

We know that Phase retrieval in $\mathbb{R}^n$ is possible using $2n - 1$ subspaces each of any dimension less than $n - 1$ [11].

Problem 9.3. Is weak phase retrieval in $\mathbb{R}^n$ possible using $2n - 2$ (or even $2n - 3$) subspaces each of any dimension less than n .

Figure 2: Problems 9.2 and 9.3 in the source paper, page 15.

2 Definitions and cofactor notation

Let $A \in \mathbb{R}^{(2n-2) \times n}$ have rows $\phi_1^T, \dots, \phi_{2n-2}^T$. Its rows do *weak phase retrieval* if

$$|\langle x, \phi_i \rangle| = |\langle y, \phi_i \rangle| \quad (1 \leq i \leq 2n-2)$$

implies that there is $\varepsilon \in \{\pm 1\}$ such that $\text{sgn}(x_j) = \varepsilon \text{sgn}(y_j)$ whenever $x_j y_j \neq 0$. The matrix is *full spark* if every n rows are linearly independent.

For $I \subset [2n-2]$ with $|I| = n-1$, let A_I be the corresponding row submatrix and define its signed cofactor vector

$$c_I(A)_j = (-1)^{j-1} \det A_{I, [n] \setminus \{j\}}, \quad 1 \leq j \leq n.$$

Then $A_I c_I(A) = 0$. Put

$$S_I(A) = \sum_{j=1}^n c_I(A)_j^2.$$

If A is full spark, every A_I has rank $n-1$, so $S_I(A) > 0$.

3 Proof intuition

Every nontrivial magnitude ambiguity of a full-spark $(2n-2)$ -row matrix comes from a balanced partition $I \sqcup I^c$ of its rows. The difference and sum of the two ambiguous signals lie on the two kernel lines $\ker A_I$ and $\ker A_{I^c}$. Weak phase retrieval holds exactly when the unit vectors on those two lines have equal absolute value in every coordinate. Cofactor vectors give polynomial coordinates for the kernel lines, turning this geometric criterion into finitely many determinant equations.

For non-density, only one balanced partition is needed. Weak phase retrieval forces one explicit cofactor polynomial to vanish, while a simple pair of shifted coordinate blocks makes that polynomial equal to one. The weak phase retrievable locus therefore lies in the zero set of a nonzero polynomial.

4 Finite cofactor characterization

Theorem 1 (Cofactor characterization). *Let $n \geq 2$ and $A \in \mathbb{R}^{(2n-2) \times n}$. The rows of A do weak phase retrieval if and only if both of the following hold:*

- (i) A is full spark;
- (ii) for every $I \subset [2n-2]$ with $|I| = n-1$ and every $j \in [n]$,

$$c_I(A)_j^2 S_{I^c}(A) = c_{I^c}(A)_j^2 S_I(A). \quad (1)$$

Equivalently, the normalized kernel generators of A_I and A_{I^c} have identical coordinatewise absolute values for every balanced partition.

Proof. Suppose first that the rows do weak phase retrieval. A known minimal-cardinality theorem says that a weak phase retrievable family of $2n-2$ vectors in $\mathbb{R}^n$ is full spark [2, Theorem 3.7]. Fix a balanced partition and set

$$u = \frac{c_I(A)}{\sqrt{S_I(A)}}, \quad v = \frac{c_{I^c}(A)}{\sqrt{S_{I^c}(A)}}.$$

For $i \in I$, u is orthogonal to ϕ_i , while for $i \in I^c$, v is orthogonal to ϕ_i . Consequently

$$|\langle u + v, \phi_i \rangle| = |\langle u - v, \phi_i \rangle| \quad (1 \leq i \leq 2n - 2).$$

Weak phase retrieval says that all nonzero products $(u_j + v_j)(u_j - v_j) = u_j^2 - v_j^2$ have one common sign, up to reversing all signs. But

$$\sum_{j=1}^n (u_j^2 - v_j^2) = \|u\|^2 - \|v\|^2 = 0.$$

Thus every term is zero: $u_j^2 = v_j^2$ for all j . Clearing denominators gives (1).

Conversely, assume (i) and (ii), and suppose $x, y \in \mathbb{R}^n$ give the same measurement magnitudes. For each row choose a sign so that $\langle x, \phi_i \rangle = \pm \langle y, \phi_i \rangle$, assigning an arbitrary sign if both inner products vanish. Let I be the plus-sign indices. Then

$$A_I(x - y) = 0, \quad A_{I^c}(x + y) = 0.$$

If $|I| \geq n$, full spark gives $x = y$; if $|I^c| \geq n$, it gives $x = -y$. The only other possibility is $|I| = |I^c| = n - 1$.

In the balanced case, there are scalars a, b such that

$$x - y = au, \quad x + y = bv,$$

where u and v are the normalized cofactor vectors above. Condition (ii) provides signs $d_j \in \{\pm 1\}$, arbitrary when $u_j = v_j = 0$, with $v_j = d_j u_j$. Hence

$$x_j y_j = \frac{(bv_j + au_j)(bv_j - au_j)}{4} = \frac{b^2 - a^2}{4} u_j^2.$$

All nonzero coordinate products therefore have the same sign. This is exactly weak equality or weak opposition of phase, proving the converse. $\square$

5 Algebraic strengthening of Problem 9.1

Corollary 1 (Proper algebraic locus). *For every $n \geq 2$, the set of weak phase retrievable $(2n - 2)$ -element families in $\mathbb{R}^n$ is contained in a proper real-algebraic hypersurface of $\mathbb{R}^{(2n-2)^n}$. In particular, it is nowhere dense and has Lebesgue measure zero.*

Proof. Fix $I_0 = \{1, \dots, n - 1\}$ and define the polynomial

$$F(A) = c_{I_0}(A)_n^2 S_{I_0^c}(A) - c_{I_0^c}(A)_n^2 S_{I_0}(A).$$

Theorem 1 gives $F(A) = 0$ for every weak phase retrievable A .

This polynomial is not identically zero. Let the first $n - 1$ rows of A_0 be $e_1^T, \dots, e_{n-1}^T$ and the last $n - 1$ rows be $e_2^T, \dots, e_n^T$. Up to irrelevant signs, $c_{I_0}(A_0) = e_n$ and $c_{I_0^c}(A_0) = e_1$, so $F(A_0) = 1$. The zero set of a nonzero real polynomial is closed, has empty interior, and has Lebesgue measure zero. It contains the weak phase retrievable locus and its closure, proving the claim. $\square$

Casazza and Akrami later proved the non-density requested in Problem 9.1 for all $m \geq 2n - 2$ [3, Theorem 6]. Corollary 1 is an independent strengthening for the minimal case: it gives an explicit algebraic certificate and measure-zero conclusion.

6 The published $\mathbb{R}^4$ example fails

The $n = 4$ example cited in the source comes from Botelho-Andrade, Casazza, Ghoreishi, Jose, and Tremain [2, p. 12]. Its measurement rows are

$$\Phi = \begin{bmatrix} 1 & 1 & 1 & -1 \\ -1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{bmatrix}.$$

this family fails to do weak phaseless reconstruction. Observe the vectors $x = (1, 2, 0)$ and $y = (2, 1, 0)$ satisfy $|\langle x, \phi_i \rangle| = |\langle y, \phi_i \rangle|$ however do not have the same absolute value in each coordinate.

Our next example is a frame which does weak phase retrieval but fails phaseless reconstruction. **Example:** We provide a set of six vectors in $\mathbb{R}^4$ which does weak phase retrieval in $\mathbb{R}^4$. In this case our vectors are the rows of the matrix:

$$\Phi = \begin{bmatrix} \phi_1 & 1 & 1 & 1 & -1 \\ \phi_2 & -1 & 1 & 1 & 1 \\ \phi_3 & 1 & -1 & 1 & 1 \\ \phi_4 & 1 & 1 & -1 & -1 \\ \phi_5 & 1 & -1 & 1 & -1 \\ \phi_6 & 1 & -1 & -1 & 1 \end{bmatrix}$$

Note that Φ fails to do phase retrieval as it requires seven vectors in $\mathbb{R}^4$ to do phase retrieval in $\mathbb{R}^4$. Given $x = (a_1, a_2, a_3, a_4)$, $y = (b_1, b_2, b_3, b_4)$ we assume

$$(3) \quad |\langle x, \phi_i \rangle|^2 = |\langle y, \phi_i \rangle|^2, \text{ for all } i = 1, 2, 3, 4, 5, 6.$$

Step 1: The following is the coefficient matrix obtained after expanding $|\langle x, \phi_i \rangle|^2$ for $i = 1, 2, \dots, 6$.

$$\frac{1}{2} \begin{bmatrix} E_1 & a_1a_2 & a_1a_3 & a_1a_4 & a_2a_3 & a_2a_4 & a_3a_4 & \sum_{i=1}^4 a_i^2 \\ E_2 & 1 & 1 & -1 & 1 & -1 & -1 & \frac{1}{2} \\ E_3 & -1 & -1 & -1 & 1 & 1 & 1 & \frac{1}{2} \\ E_4 & -1 & 1 & 1 & -1 & -1 & 1 & \frac{1}{2} \\ E_5 & 1 & -1 & -1 & -1 & -1 & 1 & \frac{1}{2} \\ E_6 & -1 & 1 & -1 & -1 & 1 & -1 & \frac{1}{2} \\ E_6 & -1 & -1 & 1 & 1 & -1 & -1 & \frac{1}{2} \end{bmatrix}$$

Figure 3: The claimed six-vector $\mathbb{R}^4$ example in arXiv:1612.08018, page 12.

Proposition 1. *The rows of Φ do not weak phase retrieval in $\mathbb{R}^4$.*

Proof. Take

$$x = (2, 1, 0, 1), \quad y = (0, -1, 0, 1).$$

Direct multiplication gives

$$|\Phi x| = |\Phi y| = (2, 0, 2, 2, 0, 2)^T.$$

On the two coordinates where both vectors are nonzero, $x_2y_2 = -1$ while $x_4y_4 = 1$. Thus x and y have neither weakly the same nor weakly opposite signs. This violates weak phase retrieval.

There is also an immediate consistency check: the first, second, fourth, and fifth rows satisfy

$$-\phi_1 + \phi_2 + \phi_4 + \phi_5 = 0.$$

Hence Φ is not full spark, contrary to the necessary full-spark theorem for a six-vector weak phase retrievable family in $\mathbb{R}^4$. $\square$

The explicit pair in Proposition 1 is decisive and does not depend on diagnosing any particular step of the published row reduction. It means that the sentence preceding Problem 9.2 should not cite this matrix as a verified $n = 4$ example. This packet does not prove that no other six-vector example in $\mathbb{R}^4$ exists.

7 Computational verification and bounded searches

The reusable exact checker in `code/verify_cofactor_theorem.py`:

- verifies full spark and every cofactor equation for the published $n = 3$ example;
- verifies the equal-magnitude counterexample to the published $n = 4$ matrix;
- checks $F(A_0) = 1$ exactly for $2 \leq n \leq 8$.

Run it from the packet directory with

```
conda run --no-capture-output -n sandbox python \
code/verify_cofactor_theorem.py
```

All checks pass using integer arithmetic. The symbolic proof above, not this finite check, establishes the theorem for every n .

The separate exact script `code/search_sign_frames_n5.py` exhausts all $\binom{16}{8} = 12,870$ projective choices of eight $\{\pm 1\}$ rows in $\mathbb{R}^5$. None is full spark. This rules out only the direct sign-vector extension of the $n = 3$ construction; it is not evidence sufficient to settle Problem 9.2.

8 Limitations and novelty check

Problem 9.2 (existence for $n \geq 5$) and Problem 9.3 (retrieval by fewer subspaces) remain open in this packet. Numerical searches in $n = 4, 5$ repeatedly approached the cofactor equations only by collapsing a coordinate or losing a minor. Those observations are recorded in the attempt log and are not used as proof.

The bounded novelty audit searched the run’s parsed arXiv corpus, cheap result indexes, and current arXiv/web results for combinations of “weak phase retrieval” with “cofactor”, “algebraic variety”, “measure zero”, “Lebesgue null”, and “nowhere dense”. It directly inspected arXiv:1612.08018, 2110.06868, 2301.03520, and 2301.05045. The search found the later non-density theorem [3], but no cofactor characterization, algebraic-nullity strengthening, or published correction of the six-vector $\mathbb{R}^4$ matrix. Novelty confidence is moderate rather than definitive because this was a bounded search, not a complete citation review.

9 Human-review recommendation

Review is recommended, with priority on three short points:

1. the sufficiency direction of Theorem 1, especially the balanced partition case;
2. the nonzero-polynomial witness in Corollary 1;
3. the direct multiplication in Proposition 1.

If these pass, the characterization and algebraic-nullity theorem should be treated as a likely new strengthening, while the $\mathbb{R}^4$ calculation should be communicated as a correction to the motivating example rather than as a solution of Problem 9.2.

References

- [1] F. Akrami, P. G. Casazza, A. Rahimi, M. A. Hasankhani Fard, and B. Daraby, *A note on (weak) phase and norm retrievable Real Hilbert space frames and projections*, arXiv:2110.06868 (2021).
- [2] S. Botelho-Andrade, P. G. Casazza, D. Ghoreishi, S. Jose, and J. C. Tremain, *Weak phase retrieval and phaseless reconstruction*, arXiv:1612.08018 (2016).
- [3] P. G. Casazza and F. Akrami, *Classifying weak phase retrieval*, arXiv:2301.03520 (2023).